

Configuring a VPN Client for Secure File Transfers (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 09

Student:

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Time on Task:

20 hours, 2 minutes

Progress:

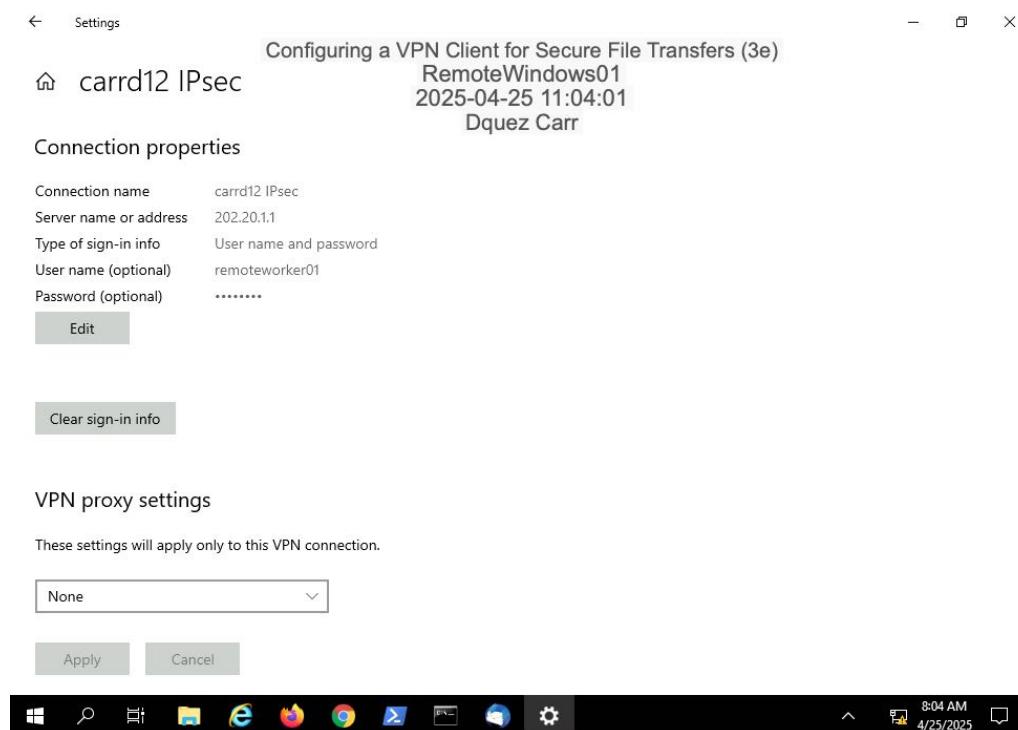
100%

Report Generated: Sunday, April 27, 2025 at 11:12 PM

Section 1: Hands-On Demonstration

Part 1: Configure a Windows VPN Client

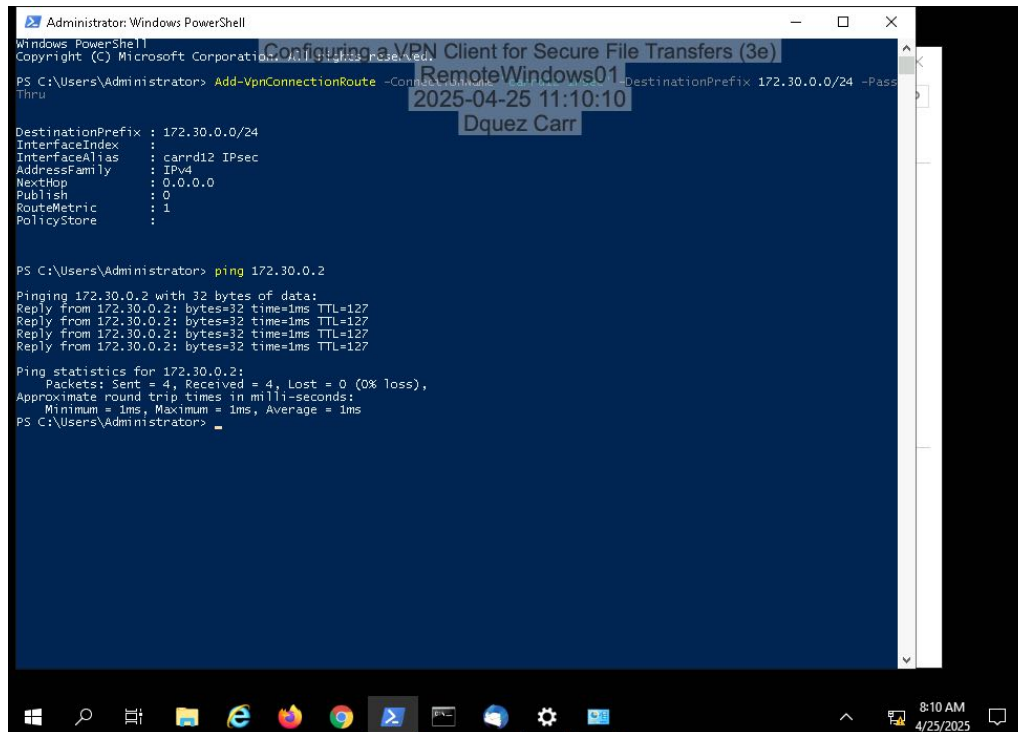
30. Make a screen capture showing the VPN connection properties.



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54. Make a screen capture showing the successful ping response.



```
Administrator: Windows PowerShell
Windows PowerShell
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PS C:\Users\Administrator> Add-VpnConnectionRoute -ConnectionName RemoteWindows01 -DestinationPrefix 172.30.0.0/24 -PassThru

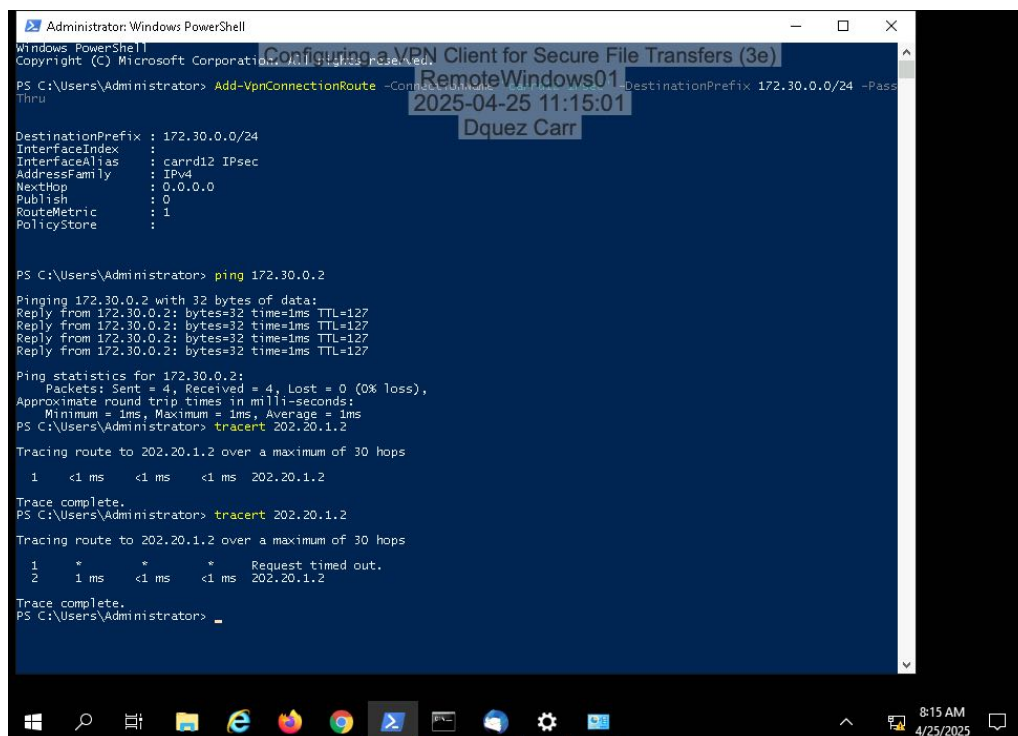
DestinationPrefix : 172.30.0.0/24
InterfaceIndex    :
InterfaceAlias    : carnd12 IPsec
AddressFamily     : IPv4
NextHop           : 0.0.0.0
Publish           : 0
RouteMetric       : 1
PolicyStore       :

PS C:\Users\Administrator> ping 172.30.0.2

Pinging 172.30.0.2 with 32 bytes of data:
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127

Ping statistics for 172.30.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
PS C:\Users\Administrator>
```

72. Make a screen capture showing your new tracert results.



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> Add-VpnConnectionRoute -ConnectionName RemoteWindows01 -DestinationPrefix 172.30.0.0/24 -PassThru

DestinationPrefix : 172.30.0.0/24
InterfaceIndex    :
InterfaceAlias    : carnd12 IPsec
AddressFamily     : IPv4
NextHop           : 0.0.0.0
Publish           : 0
RouteMetric       : 1
PolicyStore       :

PS C:\Users\Administrator> ping 172.30.0.2

Pinging 172.30.0.2 with 32 bytes of data:
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127

Ping statistics for 172.30.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
PS C:\Users\Administrator> tracert 202.20.1.2

Tracing route to 202.20.1.2 over a maximum of 30 hops
  0  <1 ms  <1 ms  <1 ms  202.20.1.2
Trace complete.
PS C:\Users\Administrator> tracert 202.20.1.2

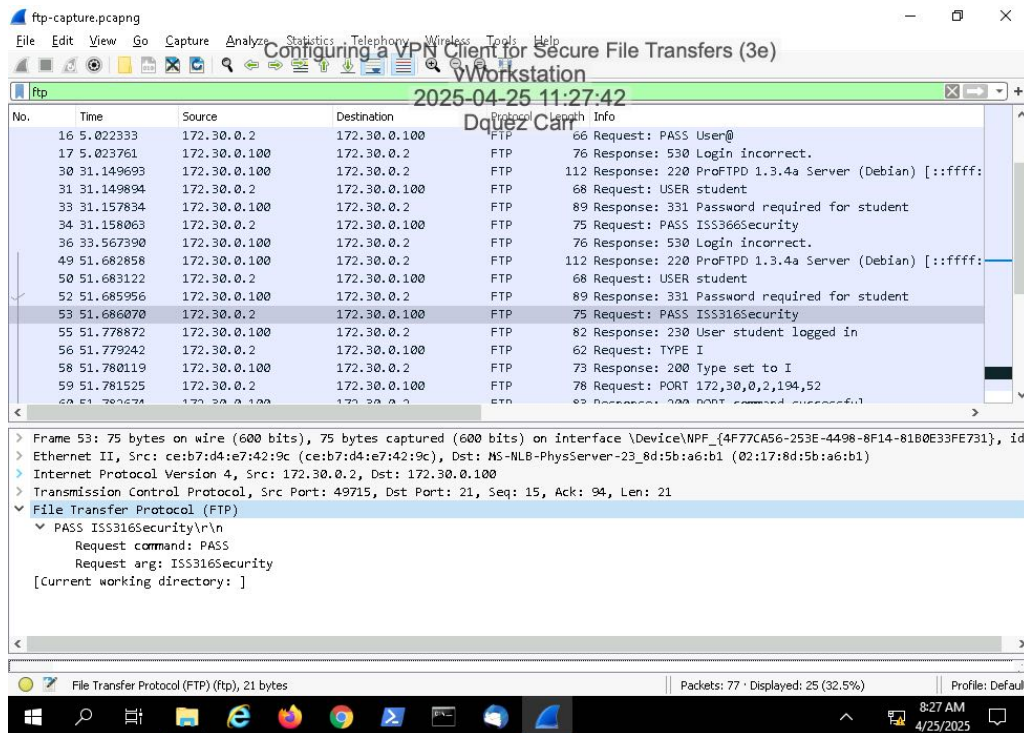
Tracing route to 202.20.1.2 over a maximum of 30 hops
  0  *      *      *      Request timed out.
  1  1 ms   <1 ms  <1 ms  202.20.1.2
Trace complete.
PS C:\Users\Administrator>
```

Part 2: Compare Secure and Non-Secure File Transfers in Wireshark

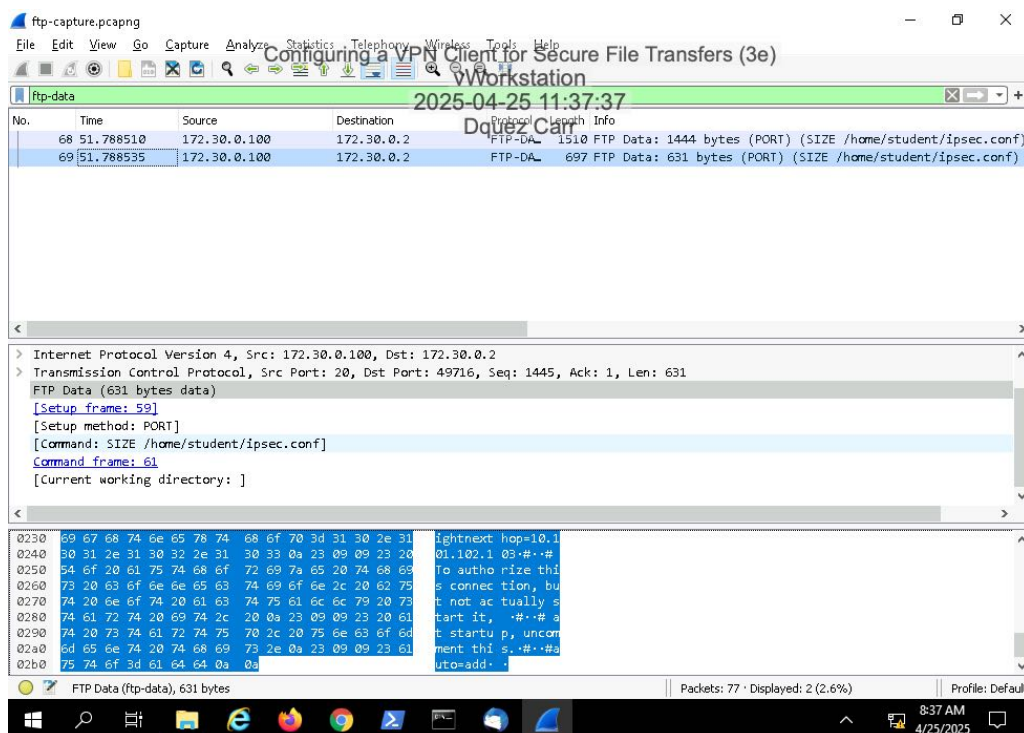
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12. Make a screen capture showing the packet that carries the correct password.



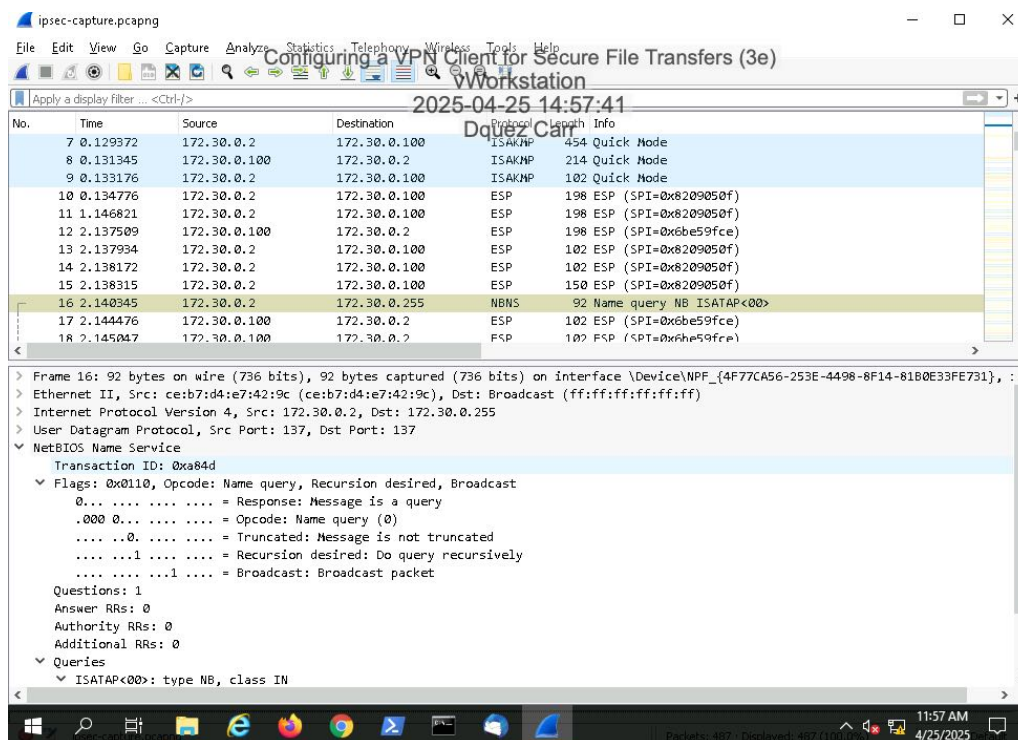
28. Make a screen capture showing the Wireshark window and the packet bytes pane for Packet 69.



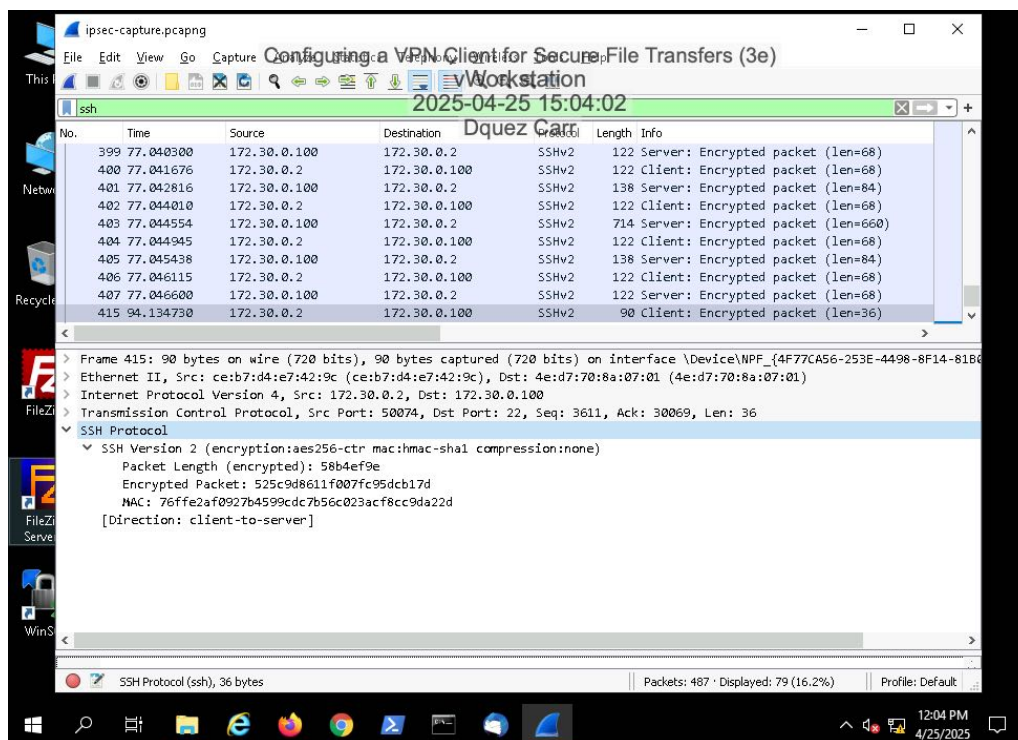
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44. Make a screen capture showing the packet details pane for packet 16.



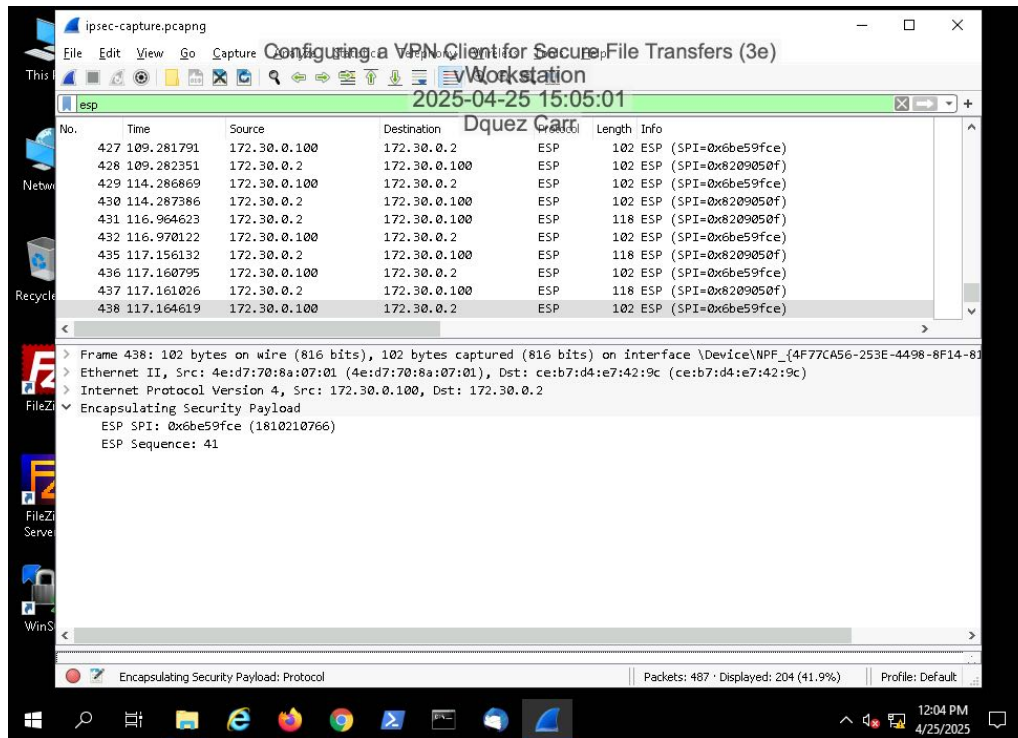
49. Make a screen capture showing the last SSHv2 packet in the SSH file transfer.



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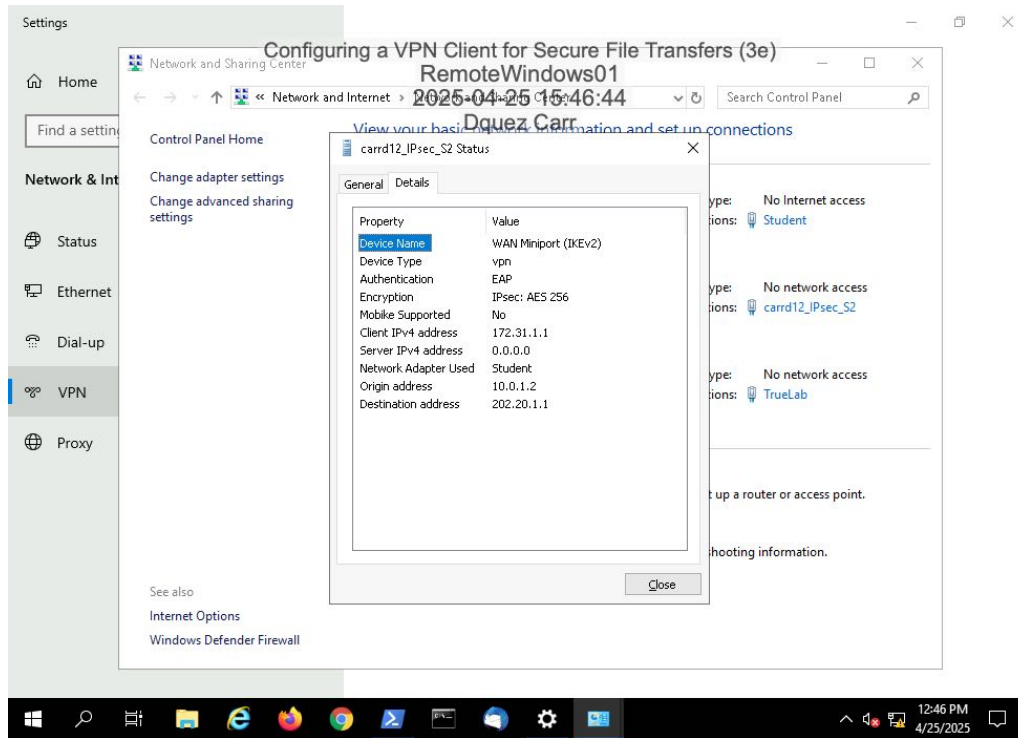
51. Make a screen capture showing the **last packets** in the ESP exchange.



Section 2: Applied Learning

Part 1: Configure a Windows VPN Client

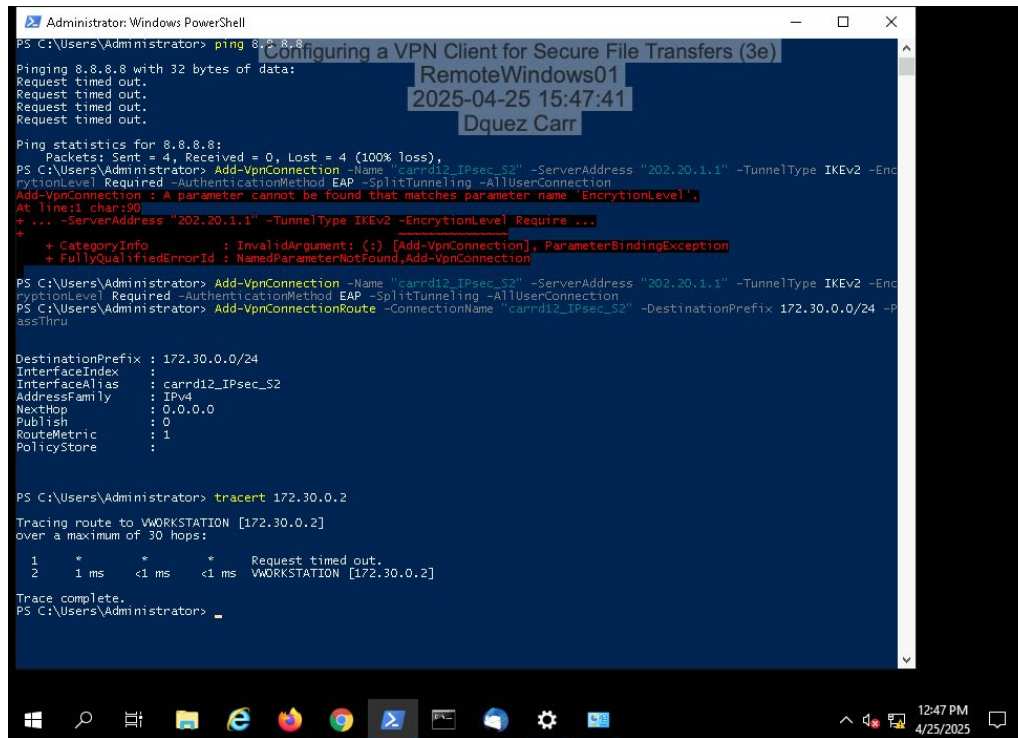
19. Make a screen capture showing the IPsec VPN connection encrypted with AES 256.



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23. Make a screen capture showing your **successful** **tracert** to the remote machine.



```
Administrator: Windows PowerShell
PS C:\Users\Administrator> ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\Administrator> Add-VpnConnection -Name "carrd12_IPsec_S2" -ServerAddress "202.20.1.1" -TunnelType IKEv2 -EncryptionLevel Required -AuthenticationMethod EAP -SplitTunneling -AllUserConnection
Add-VpnConnection : A parameter cannot be found that matches parameter name 'EncryptionLevel'.
At line:1 char:90
+ ... -ServerAddress "202.20.1.1" -TunnelType IKEv2 -EncryptionLevel Require...
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [Add-VpnConnection], ParameterBindingException
+ FullyQualifiedErrorId : NamedParameterNotFound,Add-VpnConnection

PS C:\Users\Administrator> Add-VpnConnection -Name "carrd12_IPsec_S2" -ServerAddress "202.20.1.1" -TunnelType IKEv2 -EncryptionLevel Required -AuthenticationMethod EAP -SplitTunneling -AllUserConnection
PS C:\Users\Administrator> Add-VpnConnectionRoute -ConnectionName "carrd12_IPsec_S2" -DestinationPrefix 172.30.0.0/24 -PassThru

DestinationPrefix : 172.30.0.0/24
InterfaceIndex    :
InterfaceAlias    : carrd12_IPsec_S2
AddressFamily     : IPv4
NextHop           : 0.0.0.0
Publish           : 0
RouteMetric       : 1
PolicyStore       :

PS C:\Users\Administrator> tracert 172.30.0.2

Tracing route to WORKSTATION [172.30.0.2]
over a maximum of 30 hops:
  0  * * * Request timed out.
  1  * * * Request timed out.
  2  1 ms <1 ms <1 ms WORKSTATION [172.30.0.2]

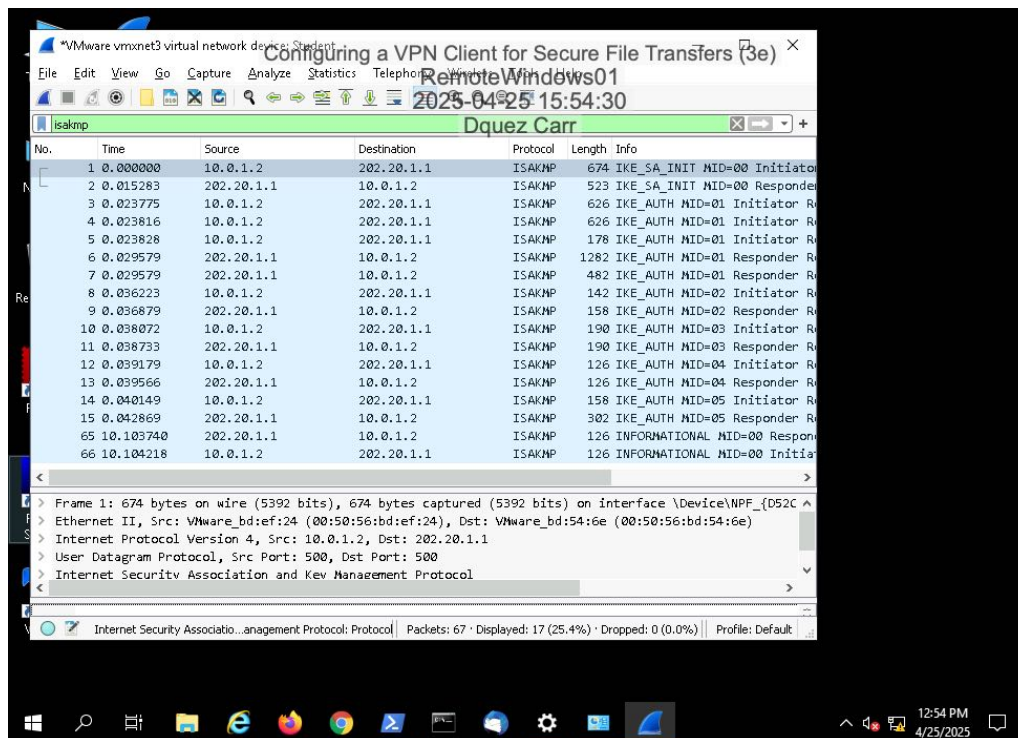
Trace complete.
PS C:\Users\Administrator>
```

Part 2: Compare Secure and Non-Secure File Transfers in Wireshark

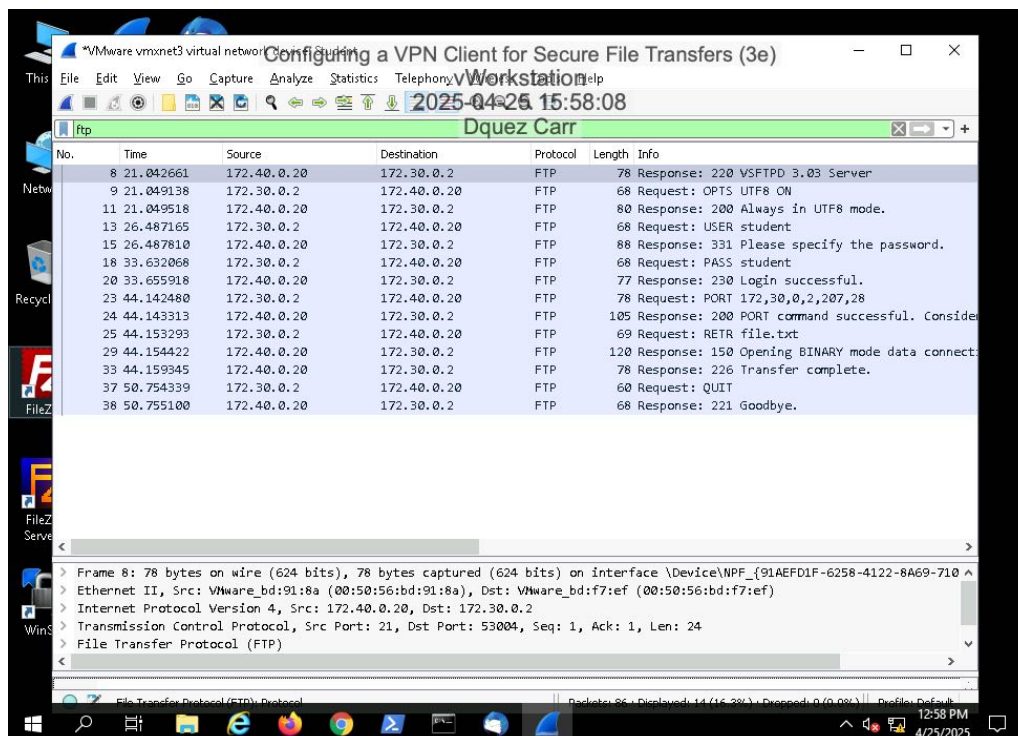
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7. Make a screen capture showing the **IKE_SA_INIT** and **IKE_AUTH** packets.



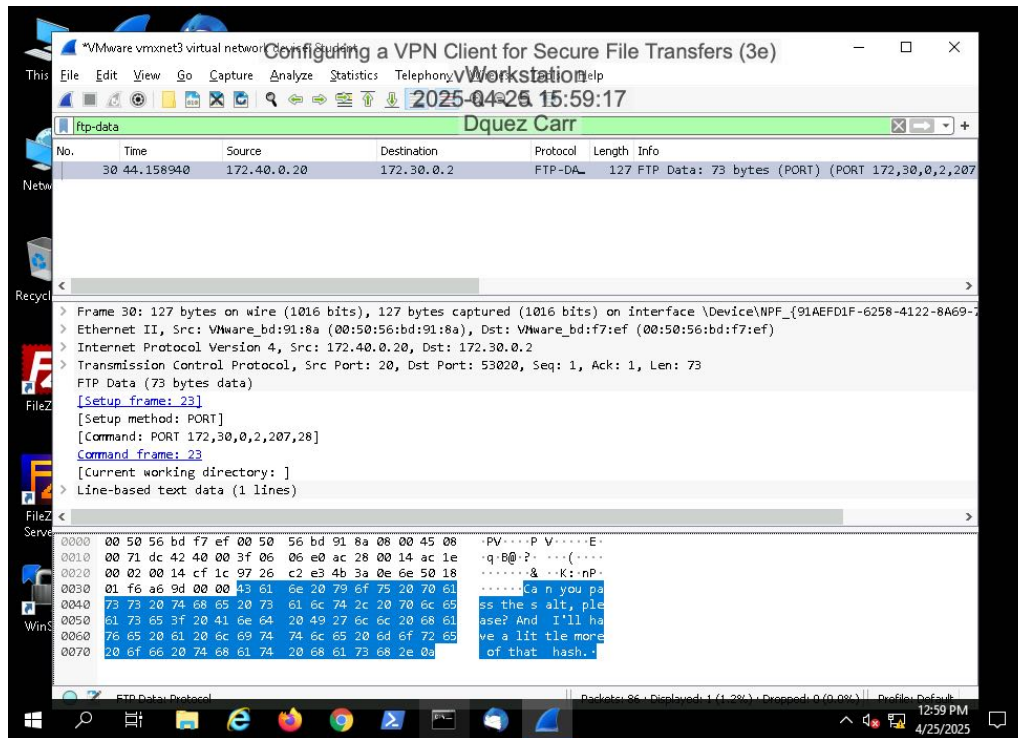
22. Make a screen capture showing the **filtered FTP packets** in your capture file.



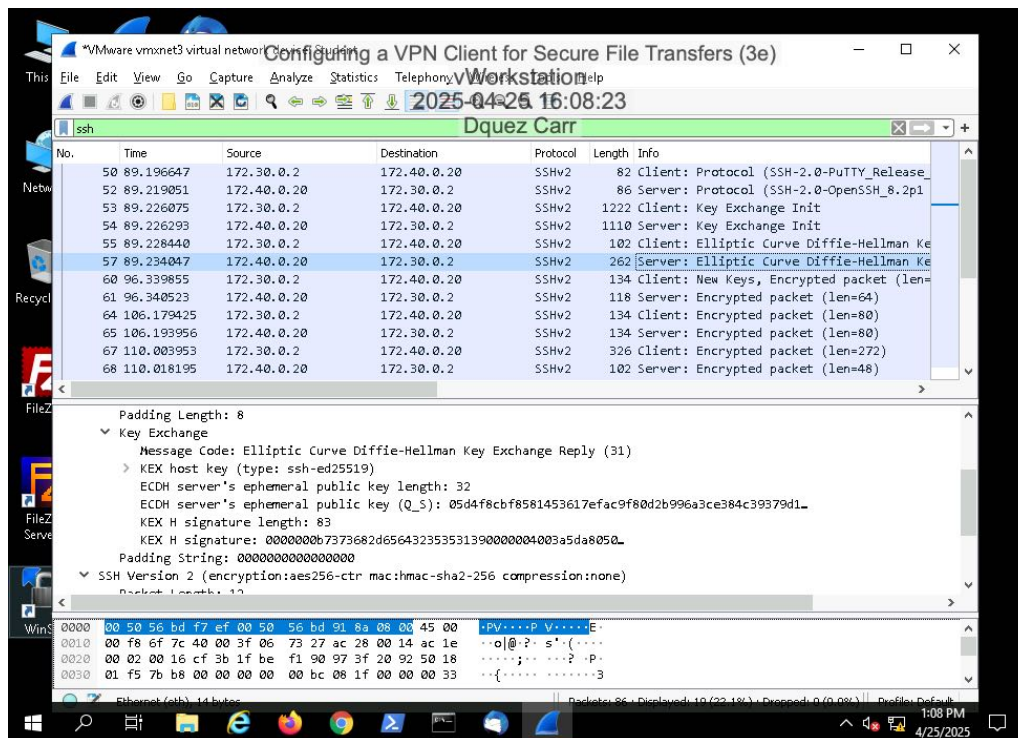
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24. Make a screen capture showing the contents of the file.txt file in the packet bytes pane.



27. Make a screen capture showing the filtered SSH packets in your capture file.



Section 3: Challenge and Analysis

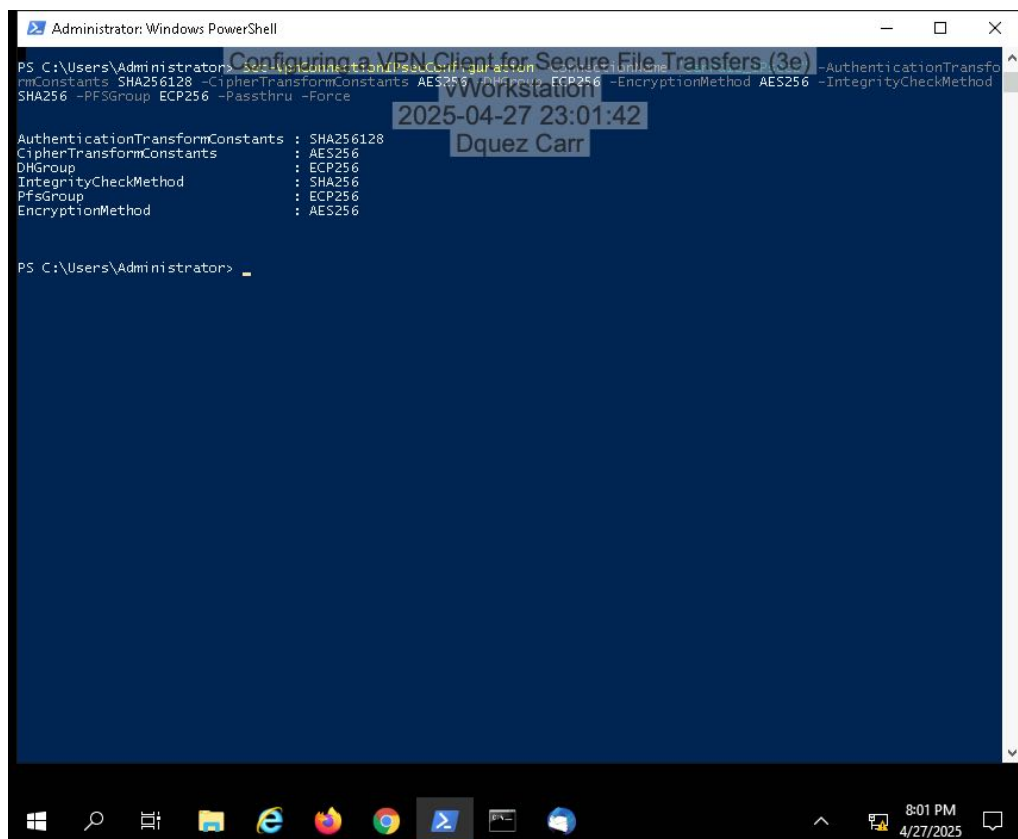
Part 1: Create a New VPN Connection using PowerShell

Document the command you used to add your VPN connection.

```
Add-VpnConnection -Name "carrd12_IPsec2" -ServerAddress "202.20.1.2" -TunnelType IKEv2  
-EncryptionLevel Required -AuthenticationMethod EAP -SplitTunneling -AllUserConnection
```

Part 2: Implement a Custom IPsec Policy

Make a screen capture showing the **successfully executed Set-VpnConnectionIPsecConfiguration** command in PowerShell.



Part 3: Verify Your VPN Implementation using Wireshark

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Make a screen capture showing the **CREATE_CHILD_SA** exchange.

VMware vmxnet3 virtual network device: Student

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

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2025-04-27 23:09:13

isakmp

No.	Time	Source	Destination	Protocol	Length	Info
251	310.195227	172.30.0.2	202.20.1.2	ISAKMP	126	INFORMATIONAL MID=63 Initiator Response
252	310.197830	202.20.1.2	172.30.0.2	ISAKMP	382	CREATE_CHILD_SA MID=64 Responder Request
253	310.202100	172.30.0.2	202.20.1.2	ISAKMP	350	CREATE_CHILD_SA MID=64 Initiator Response
255	319.911714	202.20.1.2	172.30.0.2	ISAKMP	126	INFORMATIONAL MID=65 Responder Request
256	319.912198	172.30.0.2	202.20.1.2	ISAKMP	126	INFORMATIONAL MID=65 Initiator Response
259	329.956818	202.20.1.2	172.30.0.2	ISAKMP	126	INFORMATIONAL MID=66 Responder Request
260	329.957213	172.30.0.2	202.20.1.2	ISAKMP	126	INFORMATIONAL MID=66 Initiator Response
262	331.191879	202.20.1.2	172.30.0.2	ISAKMP	126	INFORMATIONAL MID=67 Responder Request
263	331.193340	172.30.0.2	202.20.1.2	ISAKMP	126	INFORMATIONAL MID=67 Initiator Response
264	331.195749	202.20.1.2	172.30.0.2	ISAKMP	126	INFORMATIONAL MID=68 Responder Request
265	331.196170	172.30.0.2	202.20.1.2	ISAKMP	126	INFORMATIONAL MID=68 Initiator Response
266	331.198610	202.20.1.2	172.30.0.2	ISAKMP	382	CREATE_CHILD_SA MID=69 Responder Request
267	331.203206	172.30.0.2	202.20.1.2	ISAKMP	350	CREATE_CHILD_SA MID=69 Initiator Response
269	341.057439	202.20.1.2	172.30.0.2	ISAKMP	126	INFORMATIONAL MID=70 Responder Request
270	341.057895	172.30.0.2	202.20.1.2	ISAKMP	126	INFORMATIONAL MID=70 Initiator Response

< >

> UDP Encapsulation of IPsec Packets

Internet Security Association and Key Management Protocol

Initiator SPI: 8dfaa2cd460f6a77

Responder SPI: 0ea8d53047216a3f

Next payload: Encrypted and Authenticated (46)

> Version: 2.0

Exchange type: CREATE_CHILD_SA (36)

Flags: 0x00 (Responder, No higher version, Request)

.... 0... = Initiator: Responder

...0 = Version: No higher version

..0. = Response: Request

Message ID: 0x00000077

Internet Security Association and Key Management Protocol: Protocol

Packets: 274 · Displayed: 157 (57.3%)

Profile: Default

8:09 PM

4/27/2025

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Make a screen capture showing the selected Diffie-Hellman transform.

VMware vmxnet3 virtual network device: Student

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Configuring a VPN Client for Secure File Transfers (3e)

2025-04-27 23:10:32

Isakmp

No.	Time	Source	Destination	Protocol	Length	Info
306	394.202883	172.30.0.2	202.20.1.2	ISAKMP	126	INFORMATIONAL MID=82 Initiator Response
307	394.205250	202.20.1.2	172.30.0.2	ISAKMP	126	INFORMATIONAL MID=83 Responder Request
308	394.205464	172.30.0.2	202.20.1.2	ISAKMP	126	INFORMATIONAL MID=83 Initiator Response
309	394.208258	202.20.1.2	172.30.0.2	ISAKMP	382	CREATE_CHILD_SA MID=84 Responder Request
310	394.211037	172.30.0.2	202.20.1.2	ISAKMP	350	CREATE_CHILD_SA MID=84 Initiator Response
313	404.500789	202.20.1.2	172.30.0.2	ISAKMP	126	INFORMATIONAL MID=85 Responder Request
314	404.501225	172.30.0.2	202.20.1.2	ISAKMP	126	INFORMATIONAL MID=85 Initiator Response
316	414.600768	202.20.1.2	172.30.0.2	ISAKMP	126	INFORMATIONAL MID=86 Responder Request
317	414.601161	172.30.0.2	202.20.1.2	ISAKMP	126	INFORMATIONAL MID=86 Initiator Response
318	415.201745	202.20.1.2	172.30.0.2	ISAKMP	126	INFORMATIONAL MID=87 Responder Request
319	415.203871	172.30.0.2	202.20.1.2	ISAKMP	126	INFORMATIONAL MID=87 Initiator Response
320	415.206438	202.20.1.2	172.30.0.2	ISAKMP	126	INFORMATIONAL MID=88 Responder Request
321	415.206697	172.30.0.2	202.20.1.2	ISAKMP	126	INFORMATIONAL MID=88 Initiator Response
322	415.209447	202.20.1.2	172.30.0.2	ISAKMP	382	CREATE_CHILD_SA MID=89 Responder Request
323	415.213658	172.30.0.2	202.20.1.2	ISAKMP	350	CREATE_CHILD_SA MID=89 Initiator Response

...0 = Version: No higher version
...1. = Response: Response
Message ID: 0x0000004e
Length: 80
Payload: Encrypted and Authenticated (46)
Next payload: NONE / No Next Payload (0)
0... = Critical Bit: Not Critical
.000 0000 = Reserved: 0x00
Payload length: 52
Initialization Vector: e11f8020
Encrypted Data

Internet Security Association and Key Management Protocol: Protocol

Packets: 327 · Displayed: 195 (59.6%)

Profile: Default

8:10 PM
4/27/2025